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Block image encryption based on modified playfair and chaotic system

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ABSTRACT

A new image encryption was done through using chaotic theory in the diffusion and confusion processes. Diffusion process was performed by using the chaotic cross map to permutate the matrix of each image color channel. Confusion process was implemented by a modified Playfair algorithm to generalize the classical one. Playfair algorithm divides the original text of characters into blocks, where each block usually contains two-character. The main limitations of Playfair cipher are that: it encrypts only 25 characters and one cannot use special characters and numbers. In addition, keyword can only take 25 letters alone without duplicates. The new method uses two cross chaotic maps to generate 16×16 playfair matrix as a key. The image is partitioned into a definite number of blocks such that each block has 16×16 bytes. Then, each block will be encrypted by different playfair matrix. All input data such as capital letter, small letter, digits and special characters, etc. can be encrypted by this new method. Experimental results showed that the proposed playfair cipher is more secure than the classical one and very sensitive to a slight change in the secret keys. The actual results of the encryption process showed that the proposed algorithm can defend against different attacks.

1. Introduction

The developments in the information technology, communications and computations demand new forms of information security (cryptography). Asymmetric cryptography, as a kind of cryptography against the symmetric cryptography, has two keys, one for encryption and the other for decryption. While in the symmetric encryption, sender and receiver use a single shared secret key for encryption and decryption.

A playfair algorithm is one of the famous symmetric encryption methods. Researchers, at the recent decade, presented some proposals to improve the performance of Playfair algorithm. Kalaichelvi V. et al. [1] proposed an adaptive playfair cipher algorithm using Radix 64 conversion. The proposed algorithm used matrix of 8 × 8 for encryption and decryption. This matrix is padding with the given keyword called **"HACKER"** and after that the rest of the felids are padding with the remaining uppercase alphabets, lowercase alphabets, numbers (0-9) and the only two special characters (+ and /). In this system, the matrix is static and based on a known keyword, and only the lower, upper alphabet and numbers can be handled. Merzoug et al. [2] combined two types of chaotic maps (logistics and Henon map) to fill 7 × 7 matrix of Playfair. The proposed system kept the similar encryption rule, but it changed the dimension of the array of characters to 7 × 7 characters with a key of 180 bits. The matrix is filled by 26 uppercase letters, 5 special characters { ., ? ()}, numbers, 6 operators (+,−,/,*,=,%) and two special characters (#,@) to separate duplicate letters. This system is still limited to uppercase alphabets with numbers. However lowercase alphabets still cannot be handled. Basu and Kumar Ray [3] generalized the playfair cipher in which the 10×9 matrix is replaced the 5 × 5 matrix. This matrix is filling with all the uppercase and lowercase alphabets, numeric digits, other symbols as well as white space. The character ^ is used to separate the duplicate characters. However, by including the white space the cipher becomes weak as it becomes easier for the cryptanalyst to decipher the cipher text. Amalia et al. [4] encrypt text file based on combining the modified 16×16 playfair algorithm with Knapsack Naccache-Stern algorithm. They used modified Playfair cipher 16×16 algorithm to encrypt and decrypt the text file while the key is encrypted and decrypted using Knapsack Naccache-Stern algorithm. Knapsack Naccache-Stern algorithm is a robust encryption algorithm but it has a long processing time for encryption because it has three processes, which are the key generation process, encryption process, and decryption process. Hamad et al. [5] proposed a new extension of the Playfair cipher algorithm to encrypt image data. The proposed method constructs a secret key of 16×16 matrix to scramble image data

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byte by byte. The key is used to generate a mask that is subsequently XORed with the scrambled image. The results showed that using two slightly different secret keys produces completely different resultant encrypted images. Chakravarthy et al. [6] analyses 4 methods for implementing the enhanced Playfair cipher as a prerequisite step for that specific image in steganography and watermarking. The methods contain a spatial domain algorithm applied to the frequency domain of the images that give out better desired error metrics. Hardi et al. [7] implemented an image file encryption using hybrid cryptography by choosing ElGamal algorithm to perform asymmetric encryption and Double Playfair for the symmetric encryption with an acceptable running time and encrypted file size while maintaining the level of security. The stand-alone application encrypted an image with a resolution of 250 to 976 kilobytes with an acceptable running time.

This study presents a new method for image encryption by diffusion and confusion processes. The diffusion process involves chaotic cross map, while confusion process involves modifying the original Playfair algorithm based on chaotic cross map. It is believed that using chaotic cross map in the diffusion and confusion processes of the proposed algorithm can overcome the problem of high data correlation in the images.

2. Classical playfair cipher algorithm

Playfair algorithm was the first algorithm that encrypts two letters simultaneously. It depends on a 5×5 letters matrix only, which is created by entering a keyword from left to right and then filling the rest of the matrix with the remaining alphabetical letters [8]. Here, the plaintext is encrypted based on the following steps:

1. Eliminating space between words, then, dividing resulted plaintext into sections, each one has two non-similar characters. In case that any section has similar characters, the character "X" is added between them and pushing the second one to the next section. An additional "X" is added to the message end, if its length is odd.
2. Replacing two plaintext characters that placed in the same row with the characters on the right. The last character in the row is replaced circularly by the first one on the same row.
3. Replacing two plaintext characters that placed in the same column with the characters below. The last character in the column is replaced circularly by the top of the same column.
4. If the two plaintext characters are not in the same row or column, so each pair of plaintext character is replaced by the character that is in its own row and the column occupied by the other plaintext character.

For example, to encrypt the message (**COMPUTER DEPARTMENT**) by the secret keyword (**NEW WORLD**), the playfair matrix will be as in the matrix below:

N	E	W	O	R
L	D	A	B	C
F	G	H	I/J	K
M	P	Q	S	T
U	V	X	Y	Z

By applying the above playfair rules, the encrypted message (Ciphertext) will be:

Plaintext:	**CO**	**MP**	**UT**	**ER**	**DE**	**PA**	**RT**	**ME**	**NT**
Ciphertext:	**BR**	**PQ**	**ZM**	**WN**	**GD**	**QD**	**CZ**	**PN**	**RM**

The major disadvantages of the conventional Playfair algorithm are:

1. The plaintext must contain only 25(the letters I and J consider as one character) capital letters, while lowercase letters cannot be handled.
2. The decrypted text has extra characters, like the letter X that added for the text of an odd number of characters. These characters could be a part of plaintext, then it cannot be removed through the decryption operation, which makes a confusion.
3. Numbers and all other printable characters cannot be handled. In other words, this algorithm cannot treat any sentence.
4. Whitespace in the plaintext is not considered as a character and so cannot be handled.

Here, to overcome these limitations, a modified playfair algorithm is proposing. The modified playfair algorithm, which uses a 16×16 matrix "playfair matrix", contains all the printable characters.

3. Chaotic map

The theory that studies the complex systems is a mathematical sub-discipline and called chaos theory. This means that chaos theory acting with systems that develop in time with dynamical behavior of a specific type. These systems are while unpredictable deterministic". Many writers declare that chaos is "a long-term of unpredictable behavior occurring actually in a dynamic system due to the sensitivity to initial conditions" [9,10].

Complex systems are referred to systems that contain many elements that move (so much motion). Computers are required these systems to calculate all the various possibilities, for examples weather system of earth, the water boiling behavior on a stove, birds' migratory patterns, or the diffusion of vegetation across a continent.

This study uses the cross map as a kind of chaotic map to enhance the security and decrease the complexity of algorithm calculations. The cross map is defined as following:

$$\begin{aligned} x_{i+1} &= 1 - \mu y_i^2 \\ y_{i+1} &= \cos\left(\mathrm{k}.\, cos^{-1} x_i\right),\ x,\ \mathrm{y} \in [1,\ -1] \end{aligned} \qquad (1)$$

Where μ and k are the system parameters. At μ=2 and $k = 6$, the dynamic system exhibits a large and diversity of behaviors [11].

4. The proposed modified playfair algorithm

The proposed algorithm generates 16×16 playfair matrix to encrypt and decrypt images. This matrix is based on four initial values for the two cross chaotic maps, and it contains all printable character. The steps of the proposed playfair matrix generation algorithm are:

1- Entering the initial values (x_0, y_0, z_0 and w_0), which must contain a precision of 10^{-16} after the floating point. These initial values are considered as the secret key for modified playfair algorithm. Cross-map dynamics behavior exhibits a great variety when its parameters have the values μ=2 and $k = 6$.

2- To remove the chaotic transient effect, two cross maps of Eq. (1) iterated 100 times and overlooks the outcomes. First cross map equation is implementing using the first two initial values (x_0, y_0) and the second cross map equation is implementing using the initial values ($x_0 = z_0$, $y_0 = w_0$).

3- Iterate two cross maps of Eq. (1) one time.

Table 1
An example of playfair matrix at initial values $x_0 = 0.7865433789023654$, $y_0 = 0.6763976254888287$, $z_0 = 0.7687262652078234$, $w_0 = 0.2655949730808071$.

	0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
0	57	36	51	246	156	236	230	59	29	233	151	162	141	3	198	164
1	171	46	90	242	18	225	16	48	173	56	109	62	185	208	169	8
2	19	37	22	223	82	197	85	221	252	166	64	84	177	106	61	253
3	80	157	226	55	182	123	234	4	204	245	70	200	153	43	196	146
4	158	34	13	97	248	79	179	184	214	126	192	206	100	53	152	176
5	194	96	116	118	110	174	49	238	95	26	132	66	232	244	104	63
6	163	243	17	73	24	75	128	195	145	218	168	114	91	81	33	94
7	131	11	23	235	210	6	159	148	137	127	52	124	31	83	117	170
8	1	183	12	14	45	161	227	108	25	134	129	205	219	149	213	201
9	99	65	180	39	0	237	191	135	207	189	77	140	102	186	160	101
10	144	44	181	76	92	193	229	138	30	178	122	78	35	38	190	111
11	154	255	58	249	240	216	93	231	172	20	47	107	89	98	60	27
12	121	209	50	239	133	247	88	74	228	187	2	115	113	199	147	142
13	136	150	28	188	250	86	54	87	10	32	68	130	251	9	143	220
14	175	165	222	21	202	203	71	167	139	67	72	211	105	5	15	69
15	41	224	125	40	103	112	7	254	212	155	241	42	215	119	217	120

4- Converting the values of (new_x, new_y, new_z and new_w) to integers by using the following equations:

$$\left.\begin{array}{l} x_1 = \text{trunc}\left((new_x + (\,new_y{*}x\ new_z{*}\ new_w))*\ 10^6\right)\ mod\ 256 \\ y_1 = \ \text{trunc}\left((new_y + (\,new_x{*}\ new_z{*}\ new_w))*\ 10^6\right)\ mod\ 256 \\ z_1 = \ \text{trunc}\left((new_z + (\,new_x{*}\ new_y{*}\ new_w))*\ 10^6\right)\ mod\ 256 \\ z_1 = \ \text{trunc}\left((new_z + (\,new_x{*}\ new_y{*}\ new_w))*\ 10^6\right)\ mod\ 256 \end{array}\right\} \tag{2}$$

Where the *trunc function* returns the integer part of the number.

5- Adding the resulted values of x_1, y_1, z_1 and w_1 to the matrix only if they are not existed in the matrix.

6- Repeating the steps 3 to 5 until the 16×16 palyfair matrix is full with numbers and no empty matrix element. Note that after every 100 iteration, the initial values are changed using the following equations:

$$\left.\begin{array}{l} x_0 = \text{frac}\ ((x_1{*}y_1)) \\ y_0 = \text{frac}\ ((y_1{*}z_1)) \\ z_0 = \text{frac}\ ((z_1{*}w_1)) \\ w_0 = \text{frac}\ ((w_1{*}x_1)) \end{array}\right\} \tag{3}$$

Where the *frac function* returns the fractional part of the number and x_1, y_1, z_1 and w_1 x_1, y_1, z_1 and w_1 are the last resulted values from two cross maps iterations.

For example, the initial values $x_0 = 0.7865433789023654$, $y_0 = 0.6763976254888287$, $z_0 = 0.7687262652078234$ and $w_0 = 0.2655949730808071$ output the playfair matrix as shown in Table 1.

The encryption method by the modified playfair algorithm is based on the following rules:

1. At first the plaintext is processed by removing all spaces and converts each character to its ASCII code. The ASCII code for space character (32 H) is added in two cases: when the message length is an odd number of characters or when plaintext letters in the same block (or section) are repeated.
2. Replacing two ASCII codes that placed in the same row of the playfair matrix with that of the code on the right and the first characters of the row circularly following the last.
3. Replacing both ASCII codes that placed in the same column with the characters below and the top characters of the column circularly following the last.
4. Other than that, replacing each a pair of ASCII codes with the code that is in its own row and the column occupied by the other ASCII codes.

For the previous example, the encryption method of the message (**COMPUTER DEPARTMENT**) by using the palyfair matrix in Table 1 is:

Plaintext	**C**	**O**	**M**	**P**	**U**	**T**	**E**	**R**	**D**	**E**	**P**	**A**	**R**	**T**	**M**	**E**	**N**	**T**
ASCII code	67	79	77	80	85	84	69	82	68	69	80	65	82	84	77	69	78	84
Cipher code	203	126	99	70	221	177	202	253	220	72	157	99	197	177	101	72	200	107
Cipher text	Ë	~	c	F	Ý	±	Ê	ý	Ü	H	□	c	Å	±	e	H	È	k

5. The proposed image encryption algorithm

To encrypt an image of any size by the proposed algorithm, the proposed algorithm implements diffusion and confusion stages.

5.1. Diffusion stage

In this stage,

1. Inputting the original bitmap image P($m \times n$), where $m \times n$ can be any size.

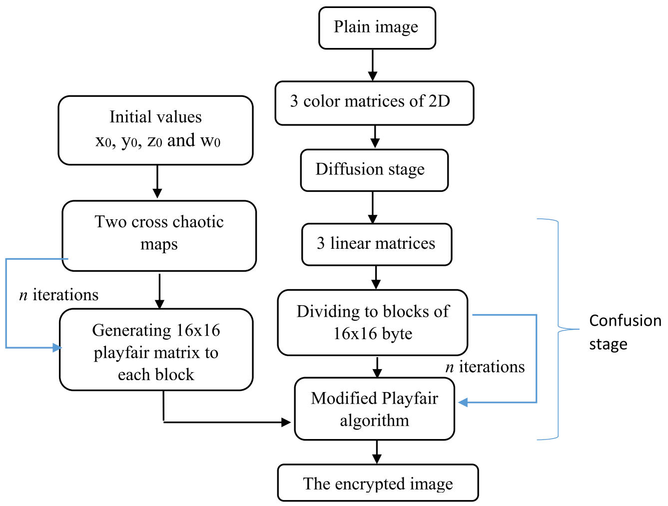

Fig. 1. The proposed encryption algorithm.

2. Separation P(mxn) into three color matrices, Pr(mxn), Pg(mxn) and Pb(mxn).
3. Generation random matrix. Two cross maps as in Eq. (1) will be iterated. This iteration will generate random numbers in range [0…no. of rows] without repetition and random numbers in range [0…no. of columns]
4. For the Pr(mxn), replacing, one by one, each pixel location by new location that generated from the previous random matrix.
5. Repeating step 4 for Pg(mxn) and Pb(mxn) after rotating the random matrix by 90 and 180 locations respectively.
6. The resulting matrices will be denoted by P′r(mxn), P′g(mxn) and P′b(mxn).

5.2. Confusion stage

Confusion stage of the proposed algorithm depends on modifying the classical Playfair algorithm. The first step in confusion stage is dividing each of P′r(mxn), P′g(mxn) and P′b(mxn) images into blocks with block size of 16×16 bytes (i.e. equal to playfair matrix size). Each block is encrypted with a different playfair matrix where each matrix is generated by different initial values. If the numbers of rows and columns of the image are not equal, the last remaining block, after dividing the plain image into (16×16 bytes) blocks, will be encrypted as it is without padding. The proposed encryption algorithm is explained in Fig. 1.

This stage contains the following steps:

1- Transforming the original color bitmap image P′r(mxn), P′g(mxn) and P′b(mxn) into P′r(mxn)x1, P′g(mxn)x1 and P′b(mxn)x1, where each one of them is a one-dimensional array.

2- For each one of these matrices the algorithm does the following:

(1) Counting the number of blocks (Bn) in original image using the following equation:

$$\text{Bn} = \text{image size div } 256 \tag{3a}$$

(2) Then checking whether the size of last block is less than 256 by using the condition: if Bn mod 256<>0 then Bn = Bn + 1

(3) The array P is divided into 16×16 byte blocks, where each block has the same size of playfair matrix.

(4) Processing the following steps for all image blocks:

a. Get the red, green and blue value of each pixel in each image block to produce three blocks bands (red block, green block and blue block).
b. Generating the playfair matrix using the initial values (x_0, y_0, z_0 and w_0).
c. Encrypt red block, green block and blue block using the modified playfair algorithm.
d. Merge the resulting three blocks to generate the output block which is inserted to the encrypted image.
e. After encrypted each block, modify the initial values (x_0,y_0,z_0 and w_0) using the following equations:

$$\left.\begin{aligned} x_0 &= \text{frac}(x_1 + y_1) \\ y_0 &= \text{frac}(y_1 + z_1) \\ z_0 &= \text{frac}(z_1 + w_1) \\ w_0 &= \text{frac}(w_1 + x_1) \end{aligned}\right\} \tag{4}$$

Where x_1, y_1, z_1 and w_1 are the last resulted values from two cross maps iterations.

(5) If the size of last block is less than 256, it will be encrypted as it is without padding.

6. Experimental results

This section presents experimental results for the proposed encryption algorithm. The proposed algorithm is implemented to encrypt bitmap color image using Delphi 6 programming language. It should be noted that the algorithm can treat any bitmap image with any size. In case of the original image is 256 × 256 color Baboon image and 206×245 color Girl image, the initial values for creating the playfair matrix for Baboon image are x_0 = 0.1768764387563808, y_0 = 0.8246763436437218, z_0 = 0.0287365797966922, w_0 = 0.0898786663348034 and the initial values for creating playfair matrix for Girl image are x_0 = 0.8798767556453391, y_0 = 0.2878721998879192, z_0 = 0.9878721990782002, w_0 = 0.5577544433676543. Fig. 2 shows the original and encrypted images. As can be seen from the

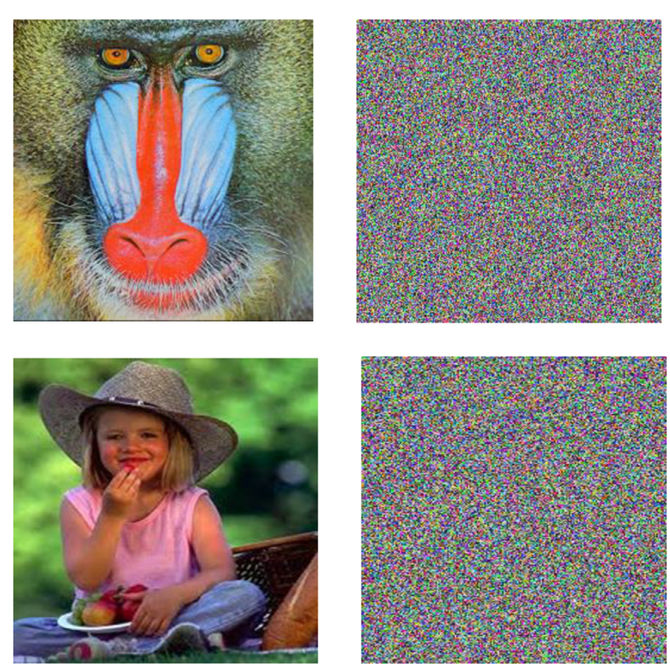

Fig. 2. (a) the original Baboon image, (b) the encrypted Baboon image, (c) original Girl image and (d) the encrypted Girl image.

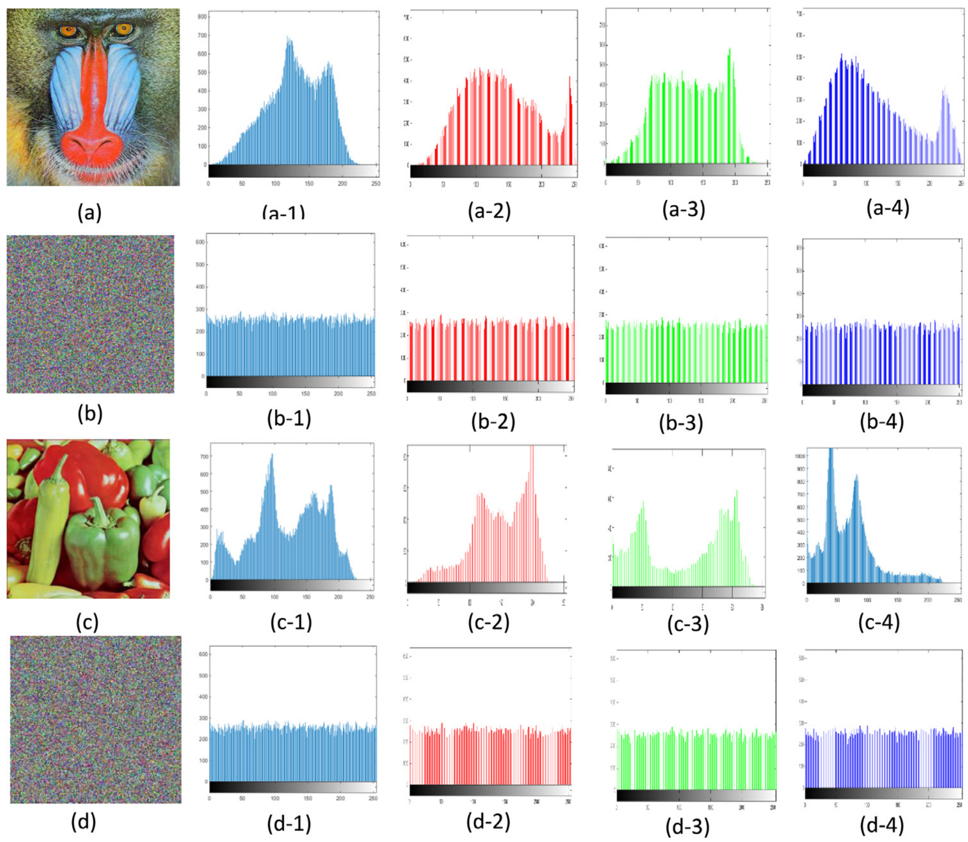

Fig. 3. Histograms analysis. (a) Baboon plain image, (a-1) histogram of (a), (a-2) histogram of Red band of (a), (a-3) histogram of Green band of (a) and (a-4) histogram of Blue band of (a). (b) Encrypted image of (a), (b-1) histogram of (b), (b-2) histogram of Red band of (b), (b-3) histogram of Green band of (b) and (b-4) histogram of Blue band of (b). (c) Peppers plain image, (c-1) histogram of (c), (c-2) histogram of Red band of (c), (c-3) histogram of Green band of (c) and (c-4) histogram of Blue band of (c). (d) Encrypted image of (c), (d-1) histogram of (d), (d-2) histogram of Red band of (d), (d-3) histogram of Green band of (d) and (d-4) histogram of Blue band of (d).

figure, there are no visible patterns or shadows in the corresponding cipher image.

7. The security and analysis

7.1. Statistical attack analysis

The statistical analysis of the original image and the encrypted image can be considering by:

7.1.1. Histogram analysis

Fig. 3 shows the histogram of the pixel numbers for any grey value in the image [2]. By taking two images a (255×255) size "Peppers" images as original images, the histogram of the original images and corresponding cipher images are shown in Fig. 3. As one see in Fig. 3, the histograms of the encrypted image and their color bands are fairly uniform, thus, no information about the plain image can be collected by analyzing the histogram because it does not offer any statistically helpful information for original image.

7.1.2. Correlation coefficient analysis

The correlation between two vertical, horizontal and diagonally adjacent pixels is analyzed for "Baboon" cipher images. 3000 pairs of contiguous pixels (vertical, horizontal, and diagonal) from plain image and its encrypted form were selected. The Eqs. (5), (6) and (7) were utilized to analyze the correlation of the contiguous pixels:

$$r_{xy} = \frac{cov(x,y)}{\sqrt{D(x)}\sqrt{D(y)}} \quad (5)$$

Table 2
Comparing correlation coefficients of two neighboring pixels between the plain and encrypted images in proposed system.

Images	Correlation of plain image			Correlation of proposed system		
	Vertical	Horizontal	Diagonal	Vertical	Horizontal	Diagonal
Baboon	0.7081	0.6018	0.7776	0.0021	−0.0159	−0.0017
Pepper	0.9472	0.9550	0.8983	0.0485	−0.0218	0.0058
Girl	0.9406	0.9276	0.9324	−0.0067	0.0013	0.0050
House	0.9753	0.9476	0.8800	−0.0146	0.0038	0.0148
Lena	0.9692	0.9856	0.9865	0.0075	−0.0055	0.0187

$$E(x) = \frac{1}{N}\sum_{i=1}^{N} x_i \quad (6)$$

$$cov(x,y) = \frac{1}{N}\sum_{i=1}^{N}(x_i - E(x))(y_{i-}E(y)) \quad (7)$$

Where cov(x,y) is the covariance, D(x) and E(x) are the variance, whereas x and y mean estimation of two contiguous pixels in the image [12]. The correlation distribution of two horizontally contiguous pixels in the two plain-image and in its cipher-image with their color bands are shown in Fig. 4.

Table 2 shows the results of correlation coefficients of 4 different images. The results illustrate that the correlation coefficients of the encrypted images are very small. These correlation analyses confirm that the chaotic encryption algorithm satisfies zero co-correlation, indicating that the attacker cannot obtain any valuable information by exploiting a statistical attack.

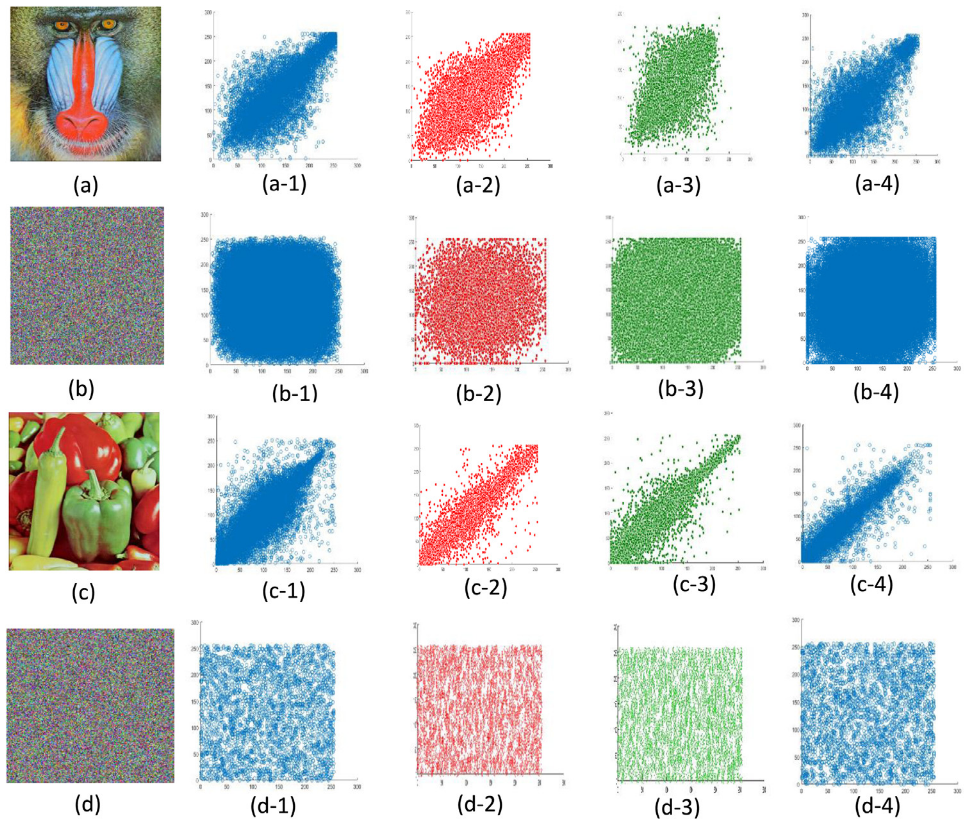

Fig. 4. Correlation analysis. (a) Baboon plain image, (a-1) Correlation of (a), (a-2) Correlation of Red band of (a), (a-3) Correlation of Green band of (a) and (a-4) Correlation of Blue band of (a). (b) Encrypted image of (a), (b-1) Correlation of (b), (b-2) Correlation of Red band of (b), (b-3) Correlation of Green band of (b) and (b-4) Correlation of Blue band of (b). (c) Peppers plain image, (c-1) Correlation of (c), (c-2) Correlation of Red band of (c), (c-3) Correlation of Green band of (c) and (c-4) Correlation of Blue band of (c). (d) Encrypted image of (c), (d-1) Correlation of (d), (d-2) Correlation of Red band of (d), (d-3) Correlation of Green band of (d) and (d-4) Correlation of Blue band of (d).

7.2. Differential attack analysis

A good cryptographic algorithm must has the ability to propagate the effect of changing in one plaintext bit as much as possible in the whole encrypted text, so that one can hide the statistical structure of plaintext in order to avoid the known plaintext attack and the chosen-plaintext attack. This means that, the taking into account the diffusion and confusion concepts in the encryption process makes the slight change in the original image produces a huge difference in the encoded image. In turn, the differential attack loses its effectiveness and turned out to be practically meaningless. The measurements examine the effect of changing just one pixel in the original image is called Number of Pixels Change Rate (NPCR) and Unified Average Changing Intensity (UACI) [13,14].

The NPCR indicator can be used to know the number of different pixels that have the same location in the plain image and in its encrypted image. The NPCR indicator is defined as following:

$$\mathrm{NPCR} = \frac{\sum_{i,j} D(i,j)}{w \text{ x } h} \text{ x } 100\% \tag{8}$$

Where, w and h are the width and height of the image, D(i, j) is defined so that D(i, j) = 0, if $C_1(i, j) = C_2(i, j)$; else D(i, j) = 1. Where $C_1(i, j)$ and $C_2(i, j)$ are two encrypted images (matrices) whose corresponding original images $I_1(i, j)$ and $I_2(i, j)$ have only one-pixel value difference.

The UACI indicator is used to know the effect on encrypted image if one pixel is changed in plain image, and it is defined as follows:

$$\mathrm{UACI} = \frac{1}{w \times h}\left(\sum_{i,j}\frac{|C_1(i,\ j) - C_2(i,\ j)|}{255}\right) \times 100\% \tag{9}$$

Table 3
Results of NPCR and UACI tests.

Images	The point	Old point value	New point value	NPCR	UACI
Baboon	(167,144)	137	233	99.1671	33.3261
Peppers	(120,150)	195	196	99.2764	33.3462
Girl	(160,18)	124	224	98.5058	32.3768
House	(15,206)	182	183	98.9049	33.5671
Lena	(200,11)	122	200	99.0127	33.6872

The ideal value of NPCR and UACI are 99.61 and 33.46 respectively [15]. UACI and NPCR measurements were used to test five color images. The results of the two indicators are close to the ideal value. The encryption algorithm is performed on the modified original image, then the results of NPCR and UACI tests are computed as shown in Table 3. The results show that a small change in the original image will result a great change in the encrypted image; this implies that the proposed algorithm has an excellent ability to resist the differential attack.

7.3. Noise attacks

The public multimedia channels such as wireless communication and internet are exposed to different types of noise and this noise would lead to distorting the quality of the images that are transmitted through these channels. This is considered a type of attack called a noise attack [16]. Shear noise, Salt and pepper noise and Gaussian noise are the common noise. The proposed algorithm has been subjected to noise attack and the performance of the algorithm is shown in Fig. 5. This experiment is done by subjecting

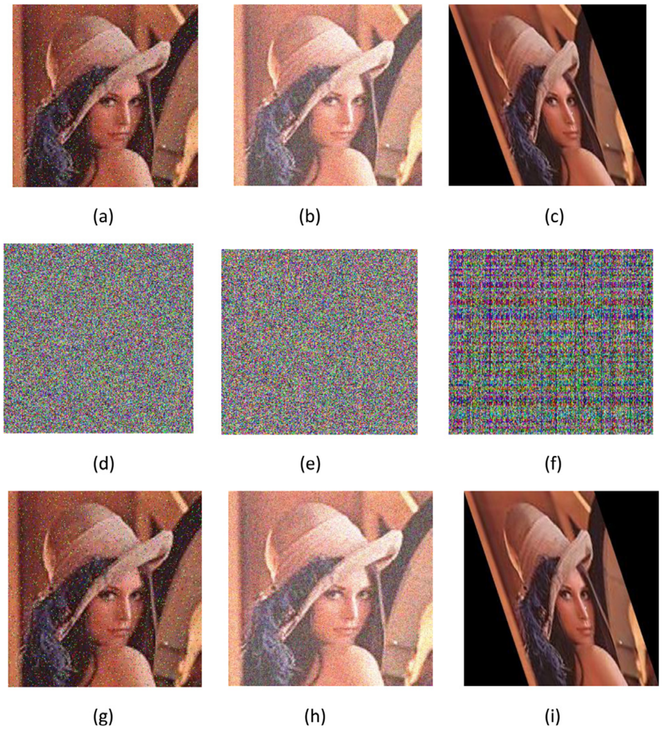

Fig. 5. Noise attacks analyses: (a)Lena image with Salt and paper noise, (b) Lena image with Gaussian noise with variance of 0.25, (c) Lena image with simple shear transformation, (d) encrypted image for (a), (e) encrypted image of (b), (f) encrypted image of (c), (g) decrypted image of (d), (h) decrypted image of (e), (i) decrypted image of (f). .

the Lena image to three types of noise: Shear noise, Salt and pepper noise and Gaussian noise as shown in the first row of Fig. 5. The second row shows the encrypted images after a Shear, Salt and pepper and Gaussian attack on a different image. The last row of Fig. 5 the images decrypted by the proposed decryption algorithm. Even though being affected by noise, these reconstructed images contain most of the original images' visual information. Despite the fact that they are affected by noise, these encrypted images contain most of the original image information in. The Mean Square Error (MSE) and PSNR between the original image and decrypted image are calculated. MSE value obtained is zero and PSNR value is infinity which means that the decrypted image is identical to original image and there is zero data loss in the decrypted image.

7.4. Information entropy analysis

Information entropy H(m) is one of the very important measurements, which is used to compute the randomness. This parameter can be computed for the plain message m as follows [13,14]:

$$H(m) = \sum_{i=1}^{N} p(m_i)\log_2 \frac{1}{p(m_i)} \tag{10}$$

Where $p(m_i)$ represents the probability mass function of the message m_i and $n = 256$ for the used image. Table 4 shows the information entropy of five different images. Every one of the entropy consequences is near the perfect estimation of the cipher entropy information, which is 8. This implies that the attacker can get less information for plain image from the distribution of the gray value in the encrypted image. So, the proposed cipher algorithm has a high security that required for the encryption algorithm. The deviation of entropy values to cipher percentage value, i.e. 8, is shown in the second row of Table 4. These deviations are small enough to give reliable encryption process.

Table 5
Results of encryption quality tests of the proposed algorithm.

Image	Maximum deviation	Irregular deviation	PSNR
Baboon	84,443	62,808	8.7897
Peppers	76,164	55,902	8.0841
Girl	79,920	43,005	8.3214
House	146,393	57,245	8.8478
Lena	120,657	56,505	8.5093

7.5. Quality of image encryption

For any encryption algorithm, pixel values are different compared to those values before encryption. So, the encryption quality will be affected by the changing rate of pixels values. This implies that encryption quality has direct connection with the aggregate changes in pixel quality between the plain image and encrypted image. The quality of image encryption can be determined as following:

1. **Maximum Deviation**: it measures the maximum deviation in plain image and cipher image. The steps of computing the maximum deviations are given in reference [17]. For this test, 5 different images were encrypted and the maximum deviation for each images pair (plain and cipher) was computed as shown in Table 5. From these results, one can observe that all images have high values of the maximum deviation. Where the greater of maximum deviation is the better to obtain high encryption quality.
2. **Irregular Deviation:** one of the good tests to judge the quality of an encryption algorithm is the image histogram, but one can't depend only on this factor alone. Another test is called the irregularity deviation that caused by encryption algorithm in the cipher image. The procedure for measuring this test is given in reference [17]. The five images pair are tested with results shown in Table 5. Better encryption quality means lower irregular value. From the obtained results, one can find that the proposed algorithm has low irregular value which indicated that it has high encryption quality.
3. **Peak Signal-to-Noise Ratio (PSNR):** it is another factor that measures the rate of pixel values between the plain image and cipher image. It can be computed by Eq. (11) [14]:

$$\mathrm{PSNR} = 10 \cdot \log_{10}\left[\frac{M \times N \times 255^2}{\sum_{i=0}^{M-1}\sum_{j=0}^{N-1}(P(i,j) - C(i,j))^2}\right] \tag{11}$$

Where M is the width and N is the height of a digital image. P(i, j) is pixel value of the plain image and C(i, j) is pixel value of the cipher image. The results of PSNR test are shown in Table 5. The good encryption algorithm must have lower PSNR value. All the resulted values in Table 5 are low values, which indicate that the proposed cipher is accepted.

Table 4
Results of entropy analysis of the proposed image cryptosystem.

	Baboon cipher image	Peppers cipher image	Girl cipher image	House cipher image	Lena cipher image
Entropy value	7.9986	7.9940	7.9930	7.9626	7.9564
Deviation percentage	0.0175%	0.075%	0.0875%	0.4675%	0.545

Table 6
comparison of Key-space, search space and frequency analysis.

	Traditional playfair	Proposed playfair	Ref. [2]	Ref. [17]
Key-space	2^{80}	2^{213}	2^{180}	2^{112}
Search-space	676	65,536	2401	4096
Frequency	0.0384	0.0039	0.0212	0.0156

7.6. Key space analysis

In order to make brute-force attacks infeasible, the proposed algorithm ought to have an extensive key space. The key space with size smaller than 2^{128} is not secured enough [18,19]. Search space must also improve because where it can be defined as "the amount of diagrams the intruder has to refer in order to figure out the relation between a given pair of pixels". This in turn facilitates in the frequency analysis of the variation of the cipher used. When the frequency of occurrence of cipher-values is lower, the security provided is greater. Here, the key space is constructed from the initial values x_0, y_0, z_0 and w_0 that are needed for creating the playfair matrix. These parameters (x_0, y_0, z_0 and w_0) represent the keys for encryption algorithm. They are floating point numbers that have values within the period [0, 1]. If the precision is 10^{-16} for each of parameters, the total space of keys for initial conditions is $((10^{16})^4) = 10^{64} \approx 2^{213}$, the search space is $256\times256 = 65{,}536$ and the frequency = 1/256=0.0039. The key space opposes sufficiently a wide expansive range of brute-force attacks. Table 6 shows the key space, search space and frequency analysis of the proposed algorithm used as compared to traditional Playfair and other implementation methods in the references [2,20].

7.7. Key sensitivity

In order to investigate the sensitivity to secret keys, the correlation coefficients between the playfair matrices, in Tables 6 and 7 with the key1 and key2, are calculated. Here, key1 is set to $x_0 = 0.7576456345435652$, $y_0 = 0.1654346688445343$, $z_0 = 0.254566767787898\underline{\mathbf{8}}$ and $w_0 = 0.4867766979776673$ and key2 is set to $x_0 = 0.7576456345435652$, $y_0 = 0.1654346688445343$, $z_0 = 0.254566767787898\underline{\mathbf{9}}$ and $w_0 = 0.4867766979776673$. The key1 is slightly changed to key2. Playfair matrices generated by key1 and key2 are shown in Tables 7 and 8, respectively, and the correlation coefficient between them is 0.0046, indicating that the S-boxes are very sensitive to the keys.

7.8. Encryption and decryption time test

In addition to the above performance factors, there is another important aspect for a good encryption algorithm, which is running speed of the algorithm. The proposed Image encryption algorithm has been implemented using Delphi 7 programming language that implemented by a computer with specification Intel core i3 6006 u @CPU 2 GHz with 4GB Ram with Win10 64bit

Table 7
playfair matrix created by using the key $x_0 = 0.7576456345435652$, $y_0 = 0.1654346688445343$, $z_0 = 0.254566767787898\underline{\mathbf{8}}$, $w_0 = 0.4867766979776673$.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	93	147	56	36	201	57	200	28	108	189	232	0	176	107	208	8
1	118	149	253	248	161	199	66	71	135	21	157	112	35	246	24	96
2	168	143	64	32	121	110	106	198	210	33	39	94	140	73	141	185
3	144	215	22	129	117	204	123	162	17	3	115	34	156	151	171	127
4	120	155	104	169	241	223	125	60	228	87	65	172	59	153	30	158
5	95	217	243	5	163	226	126	58	122	138	219	90	52	190	61	235
6	242	164	84	40	175	133	213	69	37	77	236	179	212	76	150	86
7	114	188	38	68	139	222	181	167	230	70	231	130	63	233	249	109
8	10	203	105	45	44	177	131	240	174	250	244	46	170	26	41	216
9	159	54	81	134	206	116	91	247	53	178	55	74	88	119	146	111
10	191	82	79	20	173	142	251	72	51	225	75	209	67	224	195	254
11	47	218	62	97	202	16	136	154	183	42	113	220	211	23	182	89
12	184	192	27	124	92	18	252	11	50	196	238	80	43	186	29	197
13	152	6	98	7	214	103	194	78	14	102	239	229	4	237	166	255
14	83	1	15	132	13	25	187	31	160	128	48	234	85	12	137	49
15	245	193	101	100	180	2	207	9	221	19	165	148	145	99	205	227

Table 8
Playfair matrix created using the key $x_0 = 0.7576456345435652$, $y_0 = 0.1654346688445343$, $z_0 = 0.254566767787898\underline{\mathbf{9}}$, $w_0 = 0.4867766979776673$.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	153	44	52	43	193	178	96	174	80	129	79	255	51	185	197	136
1	159	102	213	119	35	16	190	84	74	97	181	112	31	233	188	133
2	144	108	128	113	236	132	198	1	2	11	145	199	60	235	189	221
3	138	85	27	13	14	200	169	66	187	9	19	160	34	232	42	239
4	141	45	87	226	124	33	25	171	223	209	24	57	142	117	20	83
5	114	110	147	204	21	162	248	164	183	54	196	234	148	152	140	250
6	135	26	109	59	177	47	77	215	217	89	107	192	101	218	4	90
7	167	126	184	224	123	72	37	146	253	182	22	163	63	247	229	64
8	15	122	38	202	105	254	88	121	99	201	29	222	194	139	173	78
9	245	93	36	227	168	39	161	56	106	68	176	151	6	203	150	10
10	92	32	61	155	65	220	73	81	130	30	252	219	0	156	175	71
11	46	238	48	18	157	116	154	246	172	28	5	40	120	225	125	100
12	251	103	137	179	210	166	158	165	231	191	118	111	149	8	58	23
13	82	76	69	94	211	205	242	49	195	95	208	67	50	244	115	75
14	230	243	131	62	98	249	86	41	207	216	237	206	134	214	7	70
15	240	241	55	53	91	180	12	228	170	186	212	3	127	104	143	17

Table 9
Encryption time for one block and the whole image besides the decryption time for the whole image.

Image Name	No. of blocks	Encryption time for one image block in second	Encryption time for the whole image in second	Decryption time for the whole image in second
Baboon 256×256 pixels	256	0.016	0.515	0.5
Peppers 256×256 pixels	256	0.016	0.531	0.5
Girl 206×245 pixels	198	0.015	0.516	0.422
House 255×255 pixels	255	0.015	0.641	0.515
Lena 200×300 pixels	235	0.015	0.562	0.594

OS. The speed performance of an algorithm may be affected by numerous variables including the compiler utilized and the programming level. The encryption time for one block and for the whole image and the description time for the whole image was measured for 5 different images with different sizes. The analysis results for the proposed algorithm are listed in Table 9. According to these results, the encryption and decryption times have acceptable values.

8. Conclusion

This study introduces a new image encryption algorithm based on diffusion process and modified Playfair cipher to provide a high level of security. The diffusion process, which depends on the chaotic cross map, and the modified Playfair cipher, which also depends on chaotic cross map, have the main role in overcoming on the correlation in the images. The modification involved chaotic system to generate 16×16 byte playfair matrix for each block of the image. The efficiency of this method has been verified through different tests. According to implemented tests results, the proposed scheme offers high resistance against differential and statistical attacks and has a high sensitivity to any small change in the initial values of playfair matrix. The overcoming of the different tests with outstanding success leads to recommend this algorithm in secure communications.

Declaration of Competing Interest

The authors have no affiliation with any organization with a direct or indirect financial interest in the subject matter discussed in the manuscript.

CRediT authorship contribution statement

Ekhlas Abbas Albahrani: Formal analysis, Software, Writing - review & editing, Data curation, Project administration, Software, Resources. **Amal Abdulbaqi Maryoosh:** Writing - review & editing, Writing - original draft. **Sadeq H. Lafta:** Investigation, Resources, Writing - review & editing.

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